



1763
Bayes' theorem
Thomas Bayes

1920s
Factor analysis
Charles Spearman (1927)



1925–1935
Frequentist statistics
R. A. Fisher
Neyman & Pearson (1933)



1933
Probability axioms
A. N. Kolmogorov

1948
Information theory
Claude E. Shannon

1962
The Future of Data Analysis
John W. Tukey

1977
Expectation maximization
Dempster, Laird & Rubin

1997
"Statistics = Data Science?"
C. F. Jeff Wu

Late 1990s
Variational inference
Jordan, Ghahramani, Jaakkola & Saul (1999)

2015
Deep learning
LeCun, Bengio & Hinton

2017
Transformers
Vaswani et al.

2022
ChatGPT
OpenAI

PROBABILITY AND STATISTICS

DATA SCIENCE

ALGORITHMS

1795 / 1805
Least squares
Gauss / Legendre



Carl Friedrich Gauss

1901
Principal components
Karl Pearson

1946–1953
Monte Carlo & MCMC
Postwar Los Alamos
Metropolis algorithm (1953)

1945
Stored-program architecture
John von Neumann

1967
Nearest neighbours
Cover & Hart



Geoffrey Hinton

1986
Backpropagation
Rumelhart, Hinton & Williams

1980
Gigabyte disk storage
IBM 3380

1985
C++
Bjarne Stroustrup

1991
Python
Guido van Rossum

1990s–2000s
Machine learning
Support vector machines (1995)
Random forests (2001)

2012
AlexNet
ImageNet breakthrough

2006–2007
GPU computing / CUDA
NVIDIA

DATA AND COMPUTING

1936
Universal computation
Alan Turing

1970
Relational databases
E. F. Codd

1989–1991
World Wide Web
Tim Berners-Lee